ChallengeHub Final Demonstration

1. Technical Demonstration Recording Details

Course & Section	CPE106L-4 Software Design Laboratory FOPI01-01
Team Number	Team01
Team Roster	Rachel Joy Baldo (PM) | Mark Angelo Aycardo (Architect) | Jian Bernard Quinton (Git Master) | Issa Dumlao (QA Lead)
Accessible Video Link	Team01_FinalDemo_Submission.mp4
Video Duration	04:59 (4 minutes 15 seconds , strictly within 3-5 minutes boundary)
Access Verification	Public / Institutionally Accessible (Verified in incognito browser window without login prompt)

2. Timestamped Evidence Index

Timestamp	Requirement / Content Area	Video Demonstration Focus & Evidence
00:00 - 00:31	1. Problem Statement & Workflow	ChallengeHub scope, state mutation risks, and bounded workflows (UC-05 Record Activity & UC-04 View Progress).
00:31 - 01:09	2. Design Trail (Requirement to Code)	Traces FB-01/FB-04 through UML Class & Sequence diagrams (Pages 2 & 3) to Python guards in challengehub/manager.py.
01:09 - 01:44	3. Normal Path Execution	Valid 2.0 km walk log execution in Colab; record count updates 3→ 4, progress increases 7.5→9.5 km with atomic CSV write-through.
01:44 - 02:17	4. Invalid Scenario & Zero Mutation	Submits amount = 0.0; system raises ValueError: Amount must be positive and proves FB-04 zero mutation (5 →5 records).
02:17 - 03:23	5. Output Evidence & Visualization	Interprets Matplotlib cumulative progress chart and explains dynamic progress calculation via AmountProgressStrategy without stored database columns.
03:23 - 04:35	6. System Limitation & Improvement	Identifies flat-file CSV concurrency limits and proposes migrating to an embedded SQLite database with a Repository Pattern for Module IS.
04:35 - 04:59	7. Professional & Ethical Responsibility	Discusses data validity and zero-mutation equity in competitive wellness platforms.

3. Ethical AI-Assisted Use Disclosure Record

AI Tools Utilized	Google Gemini
Prompt & Scope	Script structuring to reconcile all relevant models, design, and target implementation.
Verification Method	Verified all timestamps against recorded video frames; confirmed video URL opens cleanly in incognito browser window without login credentials; executed Colab test suite.
Human Ownership	Team approved recording narration, validated execution evidence, and certified final technical accuracy.